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Deep learning aided determination of the optimal number of detectors for photoacoustic tomography

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Abstract

Photoacoustic tomography (PAT) is a non-destructive, non-ionizing, and rapidly expanding hybrid biomedical imaging technique, yet it faces challenges in obtaining clear images due to limited data from detectors or angles. As a result, the methodology suffers from significant streak artifacts and low-quality images. The integration of deep learning (DL), specifically convolutional neural networks (CNNs), has recently demonstrated powerful performance in various fields of PAT. This work introduces a post-processing-based CNN architecture named residual-dense UNet (RDUNet) to address the stride artifacts in reconstructed PA images. The framework adopts the benefits of residual and dense blocks to form high-resolution reconstructed images. The network is trained with two different types of datasets to learn the relationship between the reconstructed images and their corresponding ground truths (GTs). In the first protocol, RDUNet (identified as RDUNet I) underwent training on heterogeneous simulated images featuring three distinct phantom types. Subsequently, in the second protocol, RDUNet (referred to as RDUNet II) was trained on a heterogeneous composition of 81% simulated data and 19% experimental data. The motivation behind this is to allow the network to adapt to diverse experimental challenges. The RDUNet algorithm was validated by performing numerical and experimental studies involving single-disk, T-shape, and vasculature phantoms. The performance of this protocol was compared with the famous backprojection (BP) and the traditional UNet algorithms. This study shows that RDUNet can substantially reduce the number of detectors from 100 to 25 for simulated testing images and 30 for experimental scenarios.

1. Introduction

In the field of biomedical imaging, a variety of imaging modalities, such as tomography and microscopy, are routinely employed to accurately monitor numerous health conditions, alongside traditional spectroscopic methods. In medical practice, x-ray tomography, Magnetic Resonance Imaging (MRI), Positron Emission Tomography (PET), and ultrasound tomography are extensively used for diagnostic purposes [1, 2]. However, x-rays emit ionizing radiation, posing a threat to living cells and restricting their regular application. Similarly, MRI incorporates a powerful magnetic field ($\approx$2$-$5 T) which may pose risks for claustrophobic patients. Additionally, the use of contrast agents like Gadolinium and Dysprosium in MRI can sometimes be harmful to patients, despite providing highly detailed images [3, 4]. In PET scans, a small amount of radioactive material is injected into the body to produce images, making them unsuitable for routine checkups. Although, the ultrasound (US) technique is suitable for frequent usage but provides images with diminished clarity [5, 6].

In addition to traditional optical and acoustical techniques, the photoacoustic (PA) imaging modality has proven to be a popular choice in the biomedical field [7]. Being a hybrid modality, PA imaging involves both optical radiations to stimulate the target sample and acoustics as probing signals to acquire images of the target with a very high resolution and clarity [8]. The phenomenon of PA effect is utilized for

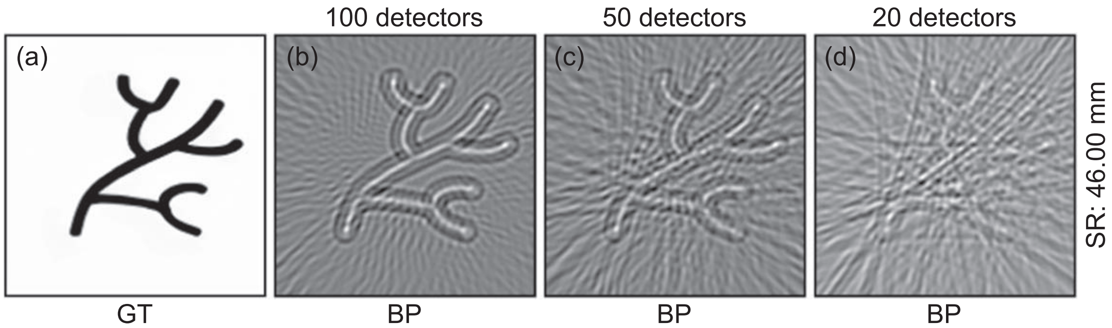

Figure 1. (a)-(d) Image reconstruction with decreasing number of detectors. The BP algorithm is used for image formation. SR: Scanning radius.

tomography imaging, referred to as the PA tomography (PAT). PAT facilitates the mapping of the medical sample by analyzing variations in the tissue's optical properties, such as absorption coefficients, which vary across different wavelengths of light, as well as the local temperature of the tissue [9]. In general, to induce the PA effect, laser pulses of nanosecond duration are used to excite the sample, and the resulting acoustic waves are detected by an ultrasound detector (USD) which rotates around the sample circularly. After data acquisition by the USD, these signals are converted into an interpretable image.

This stage involves many types of image reconstruction algorithms. There are various algorithms available, some of which are simple, known as analytical algorithms (e.g., delay and sum algorithm [10], back-projection (BP) algorithm [11], time reversal [12], etc). However, they may not generate high-quality images and could be full of artifacts, particularly when less number of detectors are used for data collection (e.g., images generated with 100 detectors certainly retain streak artifacts in full view 2D PAT). Another type is the model-based algorithm which includes system matrix [13, 14] Tikhonov regularization (e.g., l_2 and l_1 regularizations) [15], total variation [16], singular value decomposition [17], etc. Closed form solution can be obtained for the l_2 regularization method whereas the other schemes facilitate numerical solutions. It has been shown that the l_1 regularization technique can facilitate acceptable PAT images when forward data are collected using limited number of detectors (i.e., compressed/sparse sensing) [18–20]. It may be mentioned here that these algorithms, in general, may provide faithful image reconstruction even when pressure data are recorded using ≈75 detectors for full view 2D PAT. Yet, they are often computationally intensive and time-consuming. Xiang *et al* reported a 4D-PAT imaging technique essential for monitoring needle-based drug delivery and pharmacokinetics [21]. This plays a significant role in ensuring precise drug delivery to specific targets by constantly tracking the position and orientation of the needle. Moreover, the setup was effectively employed to monitor temperature changes in tissue during photothermal therapy on mouse tumors. Recently Vu *et al* proposed a GPU-based 3D reconstruction that significantly reduced the reconstruction time by five times compared with the performance of the MATLAB code [22]. These breakthroughs not only expose advanced clinical settings but also offer rapid reconstruction.

In recent years, machine learning (ML) and deep learning (DL) algorithms have gained lots of attention in the field of PAT, due to their significant performance and promising outcomes [23–27]. These ML or DL approaches are majorly categorized in two ways: pre-process and post-processing approaches. In the pre-processing approach, the PA signals are enhanced using various DL algorithms to improve the signal quality. Subsequently, a conventional reconstruction algorithm is applied to generate the final reconstructed image. Conversely, in the post-processing approach, initially, the reconstructed image is formed using a conventional reconstruction algorithm and then it is fed into a DL algorithm to further refine and enhance its quality. The UNet-based DL model is commonly used to deal with various image degradation-related problems [25, 28–30]. Various DL methods, namely, conventional UNet, dense UNet and image classification followed by dense UNet, were applied to mitigate the issue of precise measurement of scanning radius for faithful image reconstruction for full view 2D PAT [31, 32]. Many other variants of the UNet architecture can also be found in the literature [33, 34]. Notably, in most of the works, DL networks are trained and tested with similar kind of phantoms. Very few studies incorporated heterogeneous datasets, consisting of images of multiple structures/phantoms, for training and testing [29, 31, 32].

The primary objective of integrating DL into PAT is to alleviate image degradation issues caused by various algorithm-dependent,medium-dependent, and system-dependent factors. Common sources of image degradation are background noise, laser pulse width, USD spatial response, errors induced by the image reconstruction algorithm, inaccurate measurement of the scanning radius, and data sparsity. The impact of data sparsity on reconstruction quality is demonstrated in figure 1. The ground truth (GT) and corresponding reconstructed image using the BP algorithm for 100 sensors are portrayed in figures 1(a) and (b) respectively. Conversely, figures 1(c) and (d) illustrate the degradation in reconstruction quality with fewer detectors, i.e., at 50 sensors and 20 sensors. After data acquisition in any imaging modality, the most challenging task lies in converting these signals into high-quality interpretable images. According to the Shannon-Nyquist theorem, to reconstruct a signal without aliasing, the sampling frequency must be at least twice the highest-frequency component present in the signal [35]. This implies the necessity of a higher number of ultrasound detectors for reliable imaging outcomes. However, it is noted that due to the limitations of the practical experimental setup, the storage capacity of the data acquisition card, and extensive time requirements make this task complicated. As a result, it introduces streak artifacts and degrades image quality.

Antholzer *et al*, first dealt with this problem and proposed a UNet model to generate reconstructed PA images with sparse data [25]. They trained their network with simulated elliptical and similar-type of images and tested them on the simulated dataset with sparse sampling. The results show that the UNet could effectively reduce the artifacts arising from sparse sampling. Guan *et al* conducted partial view PAT imaging and employed an advanced UNet architecture referred to as the fully dense UNet for image prediction [26]. Davoudi *et al* applied the traditional UNet scheme in the context of small animal imaging [27]. They trained their network with experimental images using the PA signals captured by 512 detectors but tested with images constructed with less number of detectors e.g., 16, 32, 64, etc. Another popular deep learning framework known as the generative adversarial networks (GAN) has also been realized in the context of PAT imaging [36, 37]. The first two investigations were limited to numerical simulations only and also did not include heterogeneous datasets for training. The third study did not consider any advanced networks. Further, only experimental images were used for training and testing. Huang *et al*, used publicly available dataset for training and testing of the networks [36]. Schellenberg *et al*, explored GAN network to synthesize PAT images [37].

In this study, we proposed a post-processing DL-based convolutional neural network (CNN) model called Residual-dense UNet (RDUNet) architecture. The RDUNet model enhances feature reuse and mitigate the vanishing gradient problem. It enables RDUNet to handle complex data variations and deliver improved performance compared to other architectures [38]. RDUNet was trained in two configurations: RDUNet I, trained with simulated heterogeneous images, and RDUNet II, utilized with a combination of simulated (81%) and experimental (19%) heterogeneous datasets. Our study has four main aims: a) to systematically reduce the number of sensors and accordingly, investigate image degradation followed by DL aided image recovery, b) to find out the optimal number of sensors required to achieve acceptable image quality for each method, c) to train and test the DL networks using a heterogeneous dataset (involving images of single-disk, T-shape and vasculature phantoms) and d) to examine how RDUNet II (training set includes 81% simulated and 19% experimental images) stretches the validity domains of RDUNet I and conventional UNet (trained by the simulated images only).

2. Methods

2.1. Principle of photoacoustic tomography

2.1.1. Acoustic signal generation:
In two-dimensional PAT imaging, the target sample or medical tissue is stimulated via delta function $\delta(t)$ laser pulses from the top. As a result, this stimulation triggers the production of acoustic waves due to heating and subsequent thermoelastic expansion. The induced pressure at position $\mathbf{r}$ and time $t > 0$ is denoted by $p(\mathbf{r}, t)$ and the propagation of pressure in the acoustically homogeneous medium can be described mathematically in the following manner [15, 32]:

$$\nabla^2 p(\mathbf{r}, t) - \frac{1}{v_s^2}\frac{\partial^2 p(\mathbf{r}, t)}{\partial t^2} = -\frac{p_0(\mathbf{r})}{v_s^2}\frac{d\delta(t)}{dt}. \quad (1)$$

Here, v_s denotes the speed of sound within the medium, $p_0(\mathbf{r})$ is the initial pressure distribution and is defined as $p_0(\mathbf{r}) = \Gamma(\mathbf{r})A(\mathbf{r}) = \Gamma(\mathbf{r})\mu_a(\mathbf{r})F$, where $A(\mathbf{r})$ is the space dependent electromagnetic absorption function; $\mu_a(\mathbf{r})$ is the optical absorption coefficient; F is the fluence of the incident optical beam; $\Gamma(\mathbf{r})$ is the dimensionless Grüneisen parameter and expressed as $\Gamma(\hat{\mathbf{r}}) = v_s^2\beta/C_p$; where β stands for the isobaric volume expansion coeffricient, and, C_p represents the specific heat capacity at constant pressure. The goal of PAT reconstruction is to obtain an accurate initial pressure rise $p_0(\mathbf{r})$ using measured pressure data $p(\mathbf{r}', t)$ by the USD positioned at $\mathbf{r}'$. In reality, Γ values for various tissue media can be found in the literature [39, 40] and F can be approximately determined. Therefore, the PAT imaging essentially provides optical absorption map of the tissue under investigation.

2.1.2. PA image formation:
There are several numerical methods available to solve equation (1) for PA image formation such as the Green's function approach [41], the finite difference method [42], the finite element method [43], the pseudo-spectral method [44], etc. However, obtaining an exact analytical solution for $p_0(\mathbf{r})$ across various detection geometries poses challenges due to the complexities involved in integration or series summations [45]. To overcome this, Xu *et al* introduced a simplified time-domain reconstruction formula known as the universal BP algorithm outlined as follows [11]:

$$p_0(\mathbf{r}) = \int_{\tilde{\Omega}} b(\mathbf{r}', t)\frac{d\tilde{\Omega}}{\tilde{\Omega}}\Big|_{t=\frac{|\mathbf{r}'-\hat{\mathbf{r}}|}{v_s}}. \quad (2)$$

Here, $b(\mathbf{r}', t)$ is the BP term and expressed as [31, 46],

$$b(\mathbf{r}', t) = 2p(\mathbf{r}', t) - 2t\frac{\partial p(\mathbf{r}', t)}{\partial t}, \quad (3)$$

and $\tilde{\Omega}$ is the total solid angle subtended by the whole measurement surface and the factor $\frac{d\tilde{\Omega}}{\tilde{\Omega}}$ quantifies the

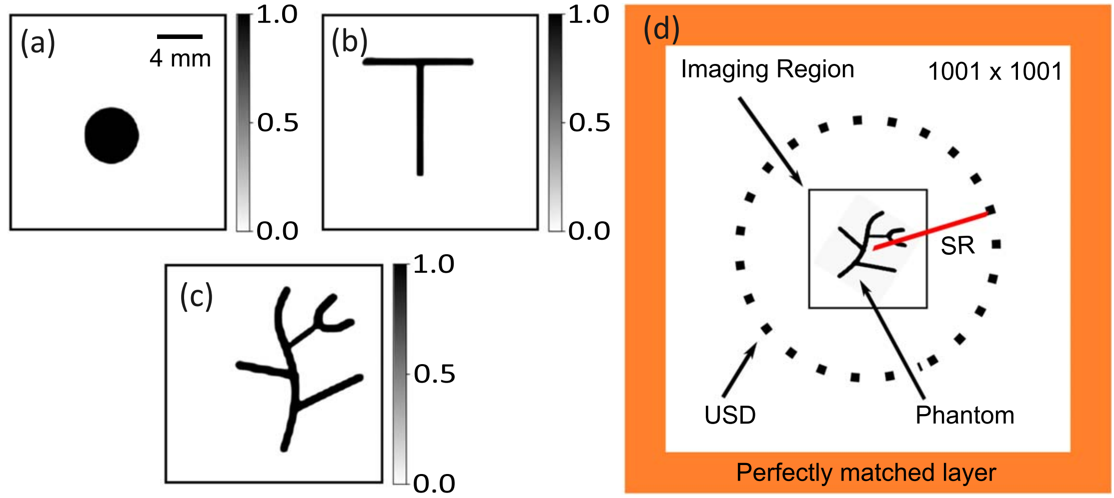

Figure 2. Demonstration of the single-disk phantom, T-shape phantom, and vasculature phantom, respectively in (a), (b), and (c) used in the numerical study. Colorbar represents the strength of the initial pressure rise. (d) The PAT setup for conducting the numerical simulations.

contribution from each element on the measurement surface.

It is essential to highlight that achieving desirable reconstruction requires capturing the PA signals from a substantial number of USD locations, and also the USD must possess a large acceptance angle to increase signal-to-noise ratio. However, practical limitations often render this scenario absurd. Consequently, the BP algorithm tends to produce low-quality images with numerous artifacts. ML or DL techniques offer promising alternatives, capable of generating higher-quality images even when PA data are captured with sparse detectors.

2.1.3. Simulation framework:
Datasets are important for DL-based algorithms to learn and enhance their prediction capabilities. The effectiveness of those algorithms strongly depends on the quality and quantity of data available. However, it is very difficult to obtain a large dataset in the medical imaging field to train a DL network. In this study, we utilized the k-Wave toolbox to generate simulated PA images [47]. The PA signals from three numerical phantoms, such as a single-disk, a T-shape structure, and a vasculature layout, were computed and accordingly, the PAT images were reconstructed. The phantoms are shown in figures 2(a), (b), & (c). These are the binary phantoms. The black pixels of a phantom absorb light energy whereas white region does not. Figure 2(d) represents the simulation setup. The computational domain was established with 1001×1001 grid points, each spaced at 0.1 mm per pixel. Additionally, a perfectly matched layer of 1 mm was applied outside of the computational domain. Ideal point detectors with a center frequency of 2.25 MHz and 70% bandwidth were evenly distributed between 0° and 360° to capture the PA signals. The simulation maintained a time interval of 20 ns for data collection, and the signal-to-noise ratio (SNR) was chosen as needed. We set the speed of sound to $v_s = 1500$ m/s and a uniform loseless acoustic medium ($\rho = 1000$ kg m^{-3}). The initial pressure value was assigned to be $p_0(\mathbf{r}) = 1$ for the black pixels and $p_0(\mathbf{r}) = 0$ for the white pixels of the imaging region. The reconstruction of PA images from the acquired PA signals was carried out using the universal BP algorithm.

2.1.4. PAT experimental procedure:
Three experimental samples were fabricated to capture the PA signals. The desired shapes were printed on transparent sheets using a desktop printer. A cylindrical base was prepared by pouring gelatin gel (8% gelatin in milk) into an acrylic cylinder. Then the printed sheet was glued to the base. The samples are displayed in figures 3 (a), (b) & (c). The black color absorbs light energy greatly whereas the surrounding region reflects the incident beam resembling the numerical phantoms.

The block diagram of the experimental setup to acquire the forward data is presented in figure 3 (d). A Q-switched Nd:YAG laser source was deployed to emit a beam of 670 nm wavelength with a repetition rate of 10 Hz and a pulse duration of 6 ns. The energy per pulse was measured to be nearly 8.9 mJ cm^{-2}, which is well behind the safety limit defined by the American National Standards Institute [48]. The laser beam traversed through three right-angled uncoated prisms (Prism 1, Prism 2, and Prism 3) and one plano-concave diffuser lens before reaching the sample. The produced PA signals were recorded using a single-element USD (V325-SU, Panametric) featuring a central frequency of 2.25MHz and a fractional bandwidth of 70%. The USD was circularly rotating around the sample during data collection at a fixed scanning radius, moving at a speed of 0. 5°/sec. Moreover, the

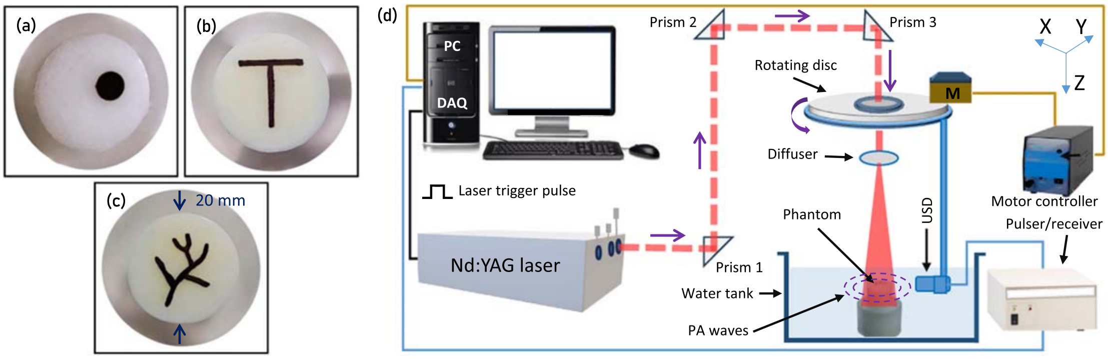

Figure 3. (a)-(c) Photographs of the phantoms. (d) Experimental setup for generation and collection of the PA signals.

Table 1. Numerical values of the parameters considered in this study to generate the datasets.

		Training data		Testing data	
SL. No.	**Parameters**	*Simulated*	*Experimental*	*Simulated*	*Experimental*
1	Scanning radius (mm)	43.00	38.00	46.00	46.00
2	No. of USD	56 to 100	56 to 100	10 to 100	10 to 100
3	USD center frequency (MHz)	2.25	2.25	2.25	2.25
4	USD bandwidth (%)	70	70	70	70
5	SNR (dB)	45	NA	35 to 55	NA
6	Image rotation	0^o to 360^o	0^o to 360^o	0^o to 360^o	0^o to 360^o
7	No. of images	$555 \times 3 = 1665$	$135 \times 3 = 405$	$19 \times 3 = 57$	$19 \times 3 = 57$

direction of propagation of light and the normal to the transducer surface is perpendicular to each other demonstrating that the PA signals were measured in the cross-direction. The detected signals were then amplified with a 55 dB gain using a pulser/receiver (DPR300, JSR Ultrasonics) and stored through a data acquisition card (PCIe-9852, ADLINK). The image reconstruction via the BP algorithm utilized the average signal (from 20 signals) for each angular location.

2.1.5. Generation of training dataset:
Two types of training datasets were generated. The first set consisted of 1665 heterogeneous reconstructed simulated images of three different kinds of phantoms ($555 \times 3 = 1665$), each paired with its GT image. These images were generated by adjusting the number of sensors ranging from 55 to 100 and having a scanning radius of 43 mm and altering the orientation. The numerical values of the relevant parameters for generating this training data are provided in table 1 (see columns 2, and 3).

On the other hand, the second dataset contained a mixture of simulated and experimental images. This amalgamation aimed to enrich the model's learning by exposing it to the real-world complexities and nuances. It included an additional 405 experimental images (135 for each type of phantom) captured at a scanning radius of 38 mm, accompanied by their respective ground truth images. In total, this dataset encompassed 2070 heterogeneous training images, consisting of 1665 simulated and 405 experimental ones. In other words, approximately 80% simulated images and 20% experimental images, all paired with their GT images, contributed to the dataset. The necessary parameters for engaged to build this dataset are provided in table 1 (see columns 2, and 4).

The reconstructed images (part of a dataset) served as inputs for various CNN algorithms, while the corresponding GT images were utilized as reference images for training the networks. The network then compared the predicted and reference images to minimize error and optimize its overall performance.

2.1.6. Generation of testing dataset:
In this study, two testing datasets were created by including simulated and experimental images. In the first dataset, 57 heterogeneous simulated reconstructed images of those three phantoms at various orientations were incorporated. The corresponding simulation setup is illustrated in section 2.1.3. These images were reconstructed using 100 sensors, with a gradual reduction in sensor counts down to 10 (i.e., 100:5:10). Ground truth images were provided for each reconstructed image. The scanning radius remained constant at 46 mm to make sure that the training and testing datasets were independent to each other. The SNR was randomly varied between 35 and 55 dB. Comprehensive details regarding the parameters are outlined in table 1 (refer to columns 2 and 5).

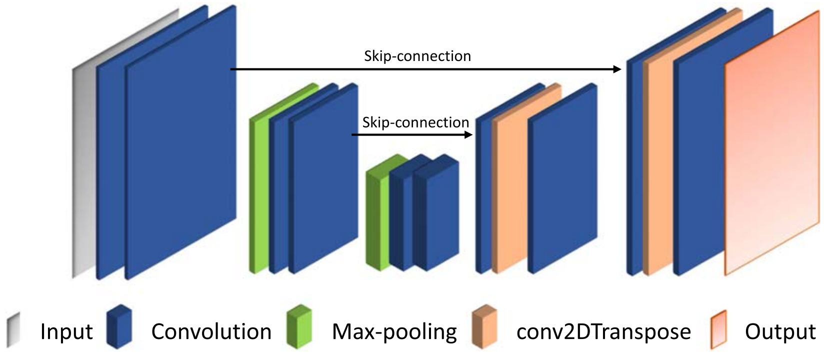

Figure 4. Embodiment of the conventional UNet architecture.

In contrast, another dataset was built involving reconstructed images formed using the PA signals provided by the custom-made setup described in section 2.1.4. It also comprised 57 heterogeneous reconstructed images for three different kinds of phantoms (19 images for each type) at different orientations. The scanning radius was fixed at approximately 46 mm. We acquired PA signals at 100 uniform locations, gradually reducing the number of detectors to 10 (uniformly placed spanning 0° to 360°). Specific information about the parameters are provided in table 1 (see columns 2 and 6).

2.2. Deep learning framework

2.2.1. Conventional UNet:

UNet-based DL networks have found extensive applications in medical imaging due to their ability to generate highly accurate imaging outcomes. It was originally investigated for semantic segmentation tasks [49]. UNet is characterized by its U-shaped architecture, which has three components: an encoder (or contracting path), a bottleneck (or bridge), and a decoder (or expanding path). Figure 4 illustrates a representative UNet architecture. The first part i.e., the encoder is responsible for capturing high-level features, gradually refining the representation of the image, and reducing the spatial dimension of the input image. It consists of successive layers of 2D convolution (conv2D) (followed by activation functions) and max-pooling operations. The bottleneck connects the encoder to the decoder. The last part, i.e., the decoder part, is opposite to the encoder; it involves a 2D transpose convolution (conv2DTranspose) operation followed by a series of conv2D operations and concatenation with feature maps from the corresponding encoder layer. This process gradually increases the spatial dimension and reconstructs the segmented output. A skip-connection combines both low-level and high-level features effectively [50]. The details of the conventional UNet architecture utilized in this study can be found elsewhere [25, 31, 49].

2.2.2. Residual dense UNet:

The detailed architecture of our proposed RDUNet has been presented in figure 5. Two major challenges in training DL networks are optimization and redundancy. To address these, we incorporated residual dense blocks in each layer of our proposed RDUNet. A detailed explanation can be found in figure 6. Residual connections enhance gradient flow and improve learning efficiency in complex architectures [38]. Furthermore, inspired by DUNet [31], we employed dense connectivity for feature reuse, effectively addressing redundancy and mitigating overfitting. Mathematically, dense connections are expressed as: $\mathbf{x}_l = H_l([\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_{l-1}])$, where $[\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_{l-1}]$ is the concatenated output of all earlier layers, and H_l represents the layer's transformation (convolution). Then a residual connection is applied across the entire block by adding the input $\mathbf{x}_0$ to its output $\mathcal{F}([\mathbf{x}_0, \mathbf{x}_1, \ldots])$, where $\mathcal{F}$ is typically 1x1 convolution, resulting in: $\mathbf{y} = \mathcal{F}([\mathbf{x}_0, \mathbf{x}_1, \ldots]) + \mathbf{x}_0$. The network took an input image of size 128×128 pixels and generated an output image of the same size.

In the encoder section, initially, the input image went through a 3×3 conv2D operation to obtain 16 feature maps from the input image of depth 1. Subsequently, it passed through a residual-based dense block. First, the dense block received an input of *f* feature maps and produced *2f* feature maps as output, with a growth rate of *f/4*. That is, it extracted 32 from 16 feature maps. Then a 3×3 conv2D operation was applied to the output of the dense block to reduce the number of channels by half, i.e., 16. That helped to keep the tensor size constant and initiated the residual connection. This output was then concatenated with the input of the dense block to complete the residual connection. Next, we applied the downsample operation to reduce the image dimension. It involved a 1×1 conv2D operation with stride 1 followed by a 3×3 conv2D operation with stride 2. It reduced the dimension of the image by half and increased the number of features by double. This process continued until the image reached the dimension of 4×4 and features increased to 512, before entering the decoder section.

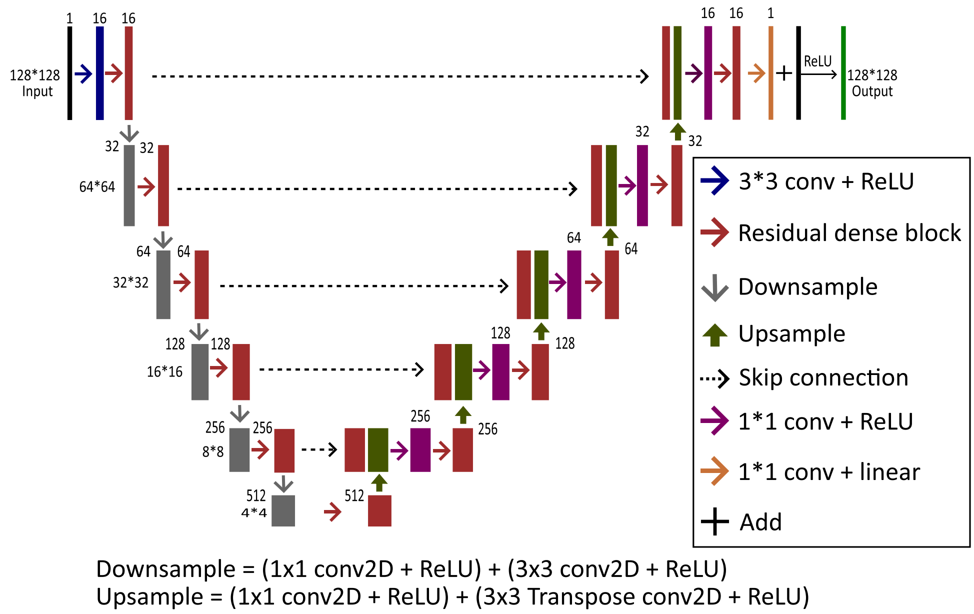

Figure 5. Building blocks of the residual dense UNet architecture.

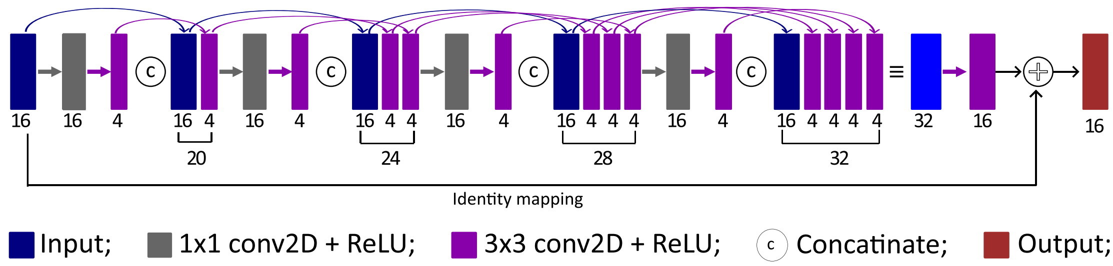

Figure 6. Illustration of the internal connections of the residual-dense block.

In the decoder section, the output of the encoder part underwent an upsample operation which worked as the opposite of a downsample operation. It helped to expand the image dimensionality by double while halving the number of features. The process was built with a 1×1 conv2D operation with stride 1 and a 3×3 conv2DTranspose operation with stride 2. As a result, the image dimension increased to 4×4 with 256 feature maps. Then a skip connection linked the output of the upsample operation with the corresponding residual-dense block in the encoder section. After that, it went through a residual-dense block to establish the residual connection. This phenomenon stretched till the image dimension reached 128×128 with 16 feature maps. Rectified linear unit (ReLU) served as a nonlinear activation function in various convolution operations. Next, it went through a 1×1 conv2D operation with linear as an activation function to produce the output image of size 128×128 with depth 1. To generate the final image, the output image was added to the input image and passed through a ReLU activation function to speed up the learning process in terms of convergence speed [51].

2.2.3. Network optimization and performance evaluation

The RDUNet was optimized using two types of datasets as detailed in the first paragraph of section 2.1.5: RDUNet I was exclusively trained using the simulated data, i.e., the first kind of dataset, whereas RDUNet II benefited from the second kind of dataset (mentioned in second paragraph of the 2.1.5), i.e., a more comprehensive dataset comprising both simulated and experimental data. After that, 90% of the images were randomly selected for training data, and the rest were used for validation data. We employed mean absolute error (MAE) as our loss function to monitor the error between predicted and actual images. For the optimization process, we applied the Adam optimizer, utilizing its default configuration parameters (learning rate $= 0.001$, beta_1 $= 0.9$, beta_2 $= 0.999$, epsilon $= 1\epsilon^{-07}$) for efficient gradient descent. Furthermore, the Glorot Normal was chosen for the kernel initializer. Both the network RDUNet I and II were trained for 100 epochs with a batch size of 8. To reduce the overfitting issue we applied a dropout of 20% in every layer. Additionally, a patience parameter of 5 was enforced during training to terminate the training process if validation

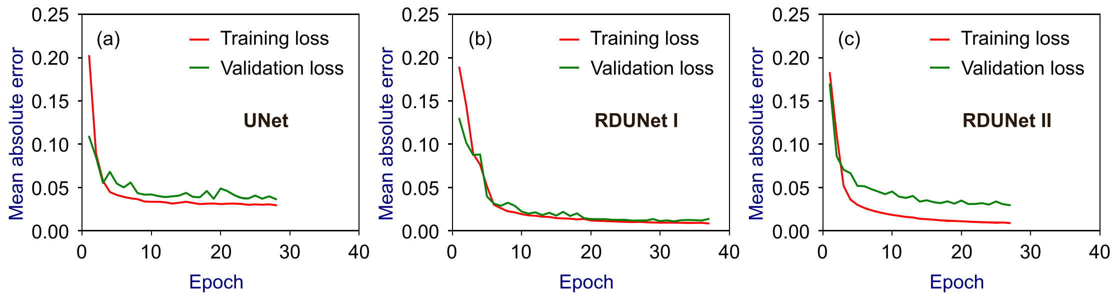

Figure 7. Graphs of mean absolute error (MAE) for different UNets during training and validation stages for each epoch.

loss was unable to improve for five consecutive epochs.

We compared the performance of RDUNet I with Antholzer's UNet architecture (details can be found in [25]). Both networks were trained on simulated images. Subsequently, we compared these results with RDUNet II (trained on the second kind of dataset). Figures 7(a), (b) and (c) illustrate the variation in MAE across epochs during the training of various DL models. For conventional UNet and RDUNet I, simulated images were used which led to convergence of training and validation loss. On the contrary, in RDUNet II, experimental images were included too. Greater variability of the dataset caused higher validation loss in later stages. Evaluation of the performance of all CNN networks was conducted on the same testing simulated and experimental datasets, as mentioned in section 2.1.6. Initially, we qualitatively analyzed the predictions of all CNNs and the reconstructions through the BP algorithm. However, the qualitative analysis varies from man to man, so we also performed quantitative comparisons. This assessment was conducted using two key metrics: the Structural Similarity Index (SSIM) and the Peak Signal-to-Noise Ratio (PSNR).

SSIM quantifies the structural similarity between the reconstructed or predicted image and the ground truth image. It is built by three main factors, namely luminance (l), contrast (c), and structure (s), and defined as follows,

$$SSIM(x, y) = [l(x, y)]^{\alpha}.[c(x, y)]^{\beta}.[s(x, y)]^{\gamma} \tag{4}$$

where,

$$l(x, y) = \frac{2\mu_x\mu_y + c_1}{\mu_x^2 + \mu_y^2 + c_1}, \quad c(x, y) = \frac{2\sigma_x\sigma_y + c_2}{\sigma_x^2 + \sigma_y^2 + c_2},$$

$$s(x, y) = \frac{\sigma_x y + c_3}{\sigma_x^2\sigma_y^2 + c_3}.$$

Here, x: GT, y: reconstructed image, μ_x, μ_y : mean, σ_x, σ_y : co-variance of the GT and reconstructed images, respectively. α,β,γ,c_1,c_2,c_3: constants.

In addition, PSNR offers a global measure of image quality, defined via the Mean Squared Error (MSE) as,

$$PSNR(x, y) = 10\log_{10}\frac{[max(x)]^2}{MSE(x, y)}. \tag{5}$$

The SSIM and PSNR values were computed using the default scikit-image library in Python while the CNNs were developed using Python version 3.9.13 and the TensorFlow v2.10 library [52, 53]. All computations were executed on a personal computer equipped with Windows 11, 64-bit architecture, 16 GB of RAM, an AMD Ryzen-5 4600H CPU, and a 4 GB NVIDIA GeForce GTX 1650 GPU.

3. Results

3.1. Simulation results

First, the performance of various CNN models was evaluated on the 19 single-disk simulated testing images. The disk was randomly positioned within the imaging domain, and reconstruction was carried out at varying noise levels ranging from 35 to 55 dB using the BP algorithm with different numbers of USDs. Figure 8 illustrates the comparative performance of the BP algorithm and the CNN networks. The GTs are shown in the first column of figure 8. Second column displays the normalized images generated by the BP algorithm (pixels values are normalized by the maximum pixel value in each image). Reconstructed images obtained via UNet, RDUNet I, and RDUNet II are inserted in third, fourth and fifth columns, respectively. Notably, the aim is to generate outputs by various methods to closely resemble the GTs. Each image is tied by a color bar indicating the numerical values of the gray levels. These images were generated using a scanning radius of 46 mm. Additionally, the numerical values of the metrics (such as SSIM and PSNR) and noise levels are also provided for analysis. The images presented in the first and second rows were generated using 50 and 25 sensors, respectively. Upon examining figures 8(b) and (g), we can conclude that the BP algorithm does not work effectively with few USDs and introduces unwanted streak artifacts in the images. The performance of the UNet architecture at 50 and 25 USDs are displayed in figures 8(c) and (h), respectively. Although UNet works well with 50 sensors; streak artifacts are surpassed but it is accompanied by ring artifacts. In the case of the 25 USDs, the reconstructed image contains

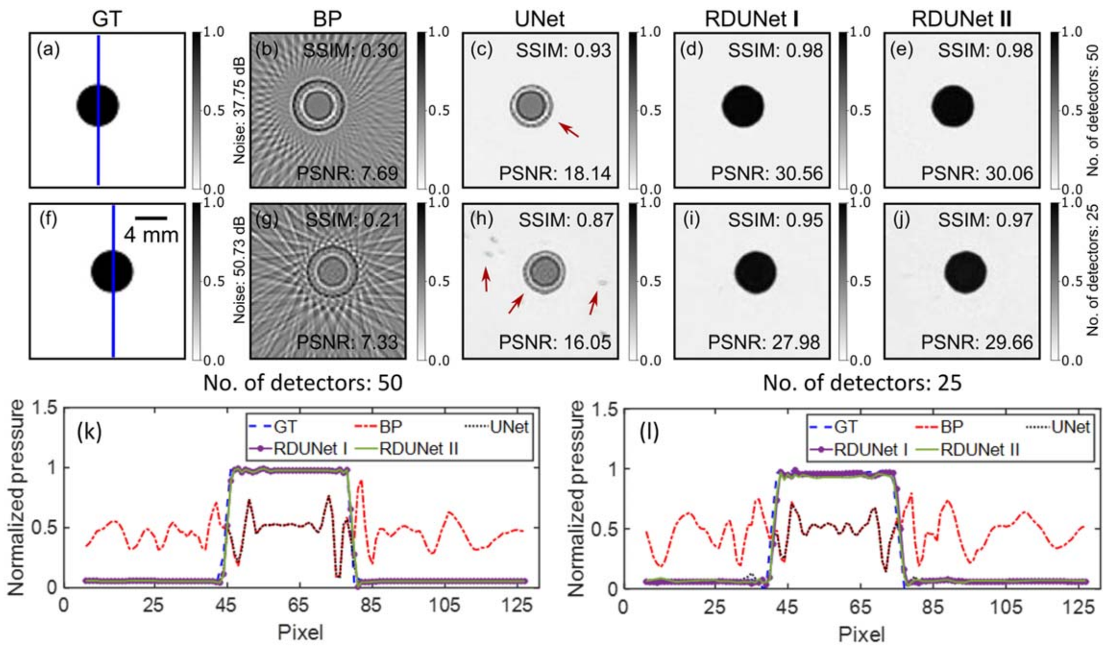

Figure 8. Recreated images by the various approaches considered in this study utilizing the simulated dataset for the single-disk phantom. (a), (f) Demonstration of the GT images. (b)-(e) The PA signals are collected by 50 uniformly spaced detectors. (g)-(j) 25 sensors have been used to capture the PA signals. The SSIM and PSNR (dB) values are inserted in each image. The red arrows indicate the unwanted information. (k), (l) Image profiles for the top and middle rows, respectively. Line plots are drawn along the blue lines as shown in (a) and (f), respectively.

some additional unwanted pieces of information. Superior performance can be observed for the RDUNet I and II. Figures 8(d) and (i) depict the performance of RDUNet I at 50 and 25 USDs, respectively, while figures 8(e) and (j) show the performance of the RDUNet II. Image profiles for the reconstructed methods are shown in figures 8(k) and (l) for the top and middle rows, respectively. The image profiles for the (b)-(e) images are drawn along the straight line presented in (a). It is clearly evident from these figures that the RDUNet I and II algorithms reproduce the image profile of the GT almost perfectly for each case. table 2 demonstrates that the average PSNR and SSIM values for the proposed methods RDUNet I and II are greater than those of other methods. The details of the SSIM and PSNR values can be found in table S1 (rows 5 to 23, columns 2 to 9). The qualitative results (i.e., correct or dispute image formation) for the CNNs are presented in table 3 for a series of blue cases (the number of USDs have been altered from 10 to 100) having noise levels randomly varying between 35 and 55 dB.

Subsequently, the reconstruction results for the T-shape phantom at 50 and 35 USDs for various algorithms are portrayed in figure 9. These images are also reconstructed using a scanning radius of 46 mm with random noise levels (48.37 and 45.93 dB). Here also, BP fails to produce acceptable reconstruction with fewer USDs. The prediction of the UNet with 50 USDs is acceptable; however, it breaks down as the number of USDs reduces, e.g., for 35 USDs. RDUNet I exhibits a similar pattern of performance, where the prediction performance deteriorates with a decreasing number of sensors. Nonetheless, RDUNet II produces almost accurate reconstructions in both cases, i.e., 50 and 35 detectors. The superior performance of the RDUNet II can also be seen form figures 9(k) and (l). The quantitative performance of the algorithms can also be observed from table 2, where RDUNet I and II outperform other algorithms. Computed values of SSIM and PSNR are inserted in table S2 (rows 5 to 23, columns 2 to 9). It can be seen from table 3 that the performance is compromised below 35 USDs.

Lastly, the reconstructed images for a more complex structure, the vasculature phantom, are presented in figure 10. Normalized BP images deviate significantly from the corresponding GTs, while various CNN models perform well with 50 USDs. However, the performance of the UNet diminishes significantly with 30 sensors, whereas RDUNet I and II exhibit close matches to the GT throughout but RDUNet I contains an unwanted mark at the bottom of the image. To compare the reconstruction schemes quantitatively, we draw image profiles and showcase in figures 10(k) and (l) for the two cases. The average values of SSIM and PSNR are provided in table 2. The quantitative values of table S3 (rows 5 to 23, columns 2 to 9) present the numerical values of SSIM and PSNR for all cases. The qualitative result is pasted in table 3 for various algorithms among which RDUNet II can provide an accurate reconstruction up to 30 USDs.

3.2. Experimental results

The favorable outcomes achieved by the RDUNet on the simulated testing dataset to improve the quality of

Table 2. Quantitative analysis to evaluate the performance of various algorithms (NA stands for not applicable; Training time: to train a network; Testing time: to produce an image).

	Average SSIM						Average PSNR (dB)								
	Simulation			*Experiment*			*Simulation*			*Experiment*					
Method	single-disk	T-shape	Vasculature	single-disk	T-shape	Vasculature	single-disk	T-shape	Vasculature	single-disk	T-shape	Vasculature	Epoch	Training time (min)	Testing time (ms)
BP	0.34	0.43	0.31	0.46	0.52	0.33	7.36	5.79	6.93	6.15	8.22	7.45	NA	NA	50
UNet	0.90	0.91	0.87	0.91	0.85	0.80	17.55	18.82	16.98	15.98	15.91	14.95	28	11	9
RDUNet I	0.93	0.94	0.94	0.93	0.85	0.87	27.89	23.98	22.42	23.82	14.90	19.21	37	18	5
RDUNet II	0.95	0.94	0.94	0.96	0.96	0.88	28.26	24.09	22.33	29.65	26.10	18.75	27	15	5

Table 3. Summary of performance of different UNet architectures; here (*d*, *c*) defective and correct image reconstruction by the conventional UNet; (*D*, C) defective and correct image creation by the RDUNet I; (**D, C**) defective and correct image formation by the RDUNet II.

	single-disk							**T-shape**							**Vasculature**						
	Simulation				*Experiment*			*Simulation*				*Experiment*			*Simulation*				*Experiment*		
No. of detectors	**Noise (dB)**	**Prediction**			**Prediction**			**Noise (dB)**	**Prediction**			**Prediction**			**Noise (dB)**	**Prediction**			**Prediction**		
10	35.18	d	*D*	**D**	d	*D*	**D**	53.34	d	*D*	**D**	d	*D*	**D**	49.86	d	*D*	**D**	d	*D*	**D**
15	39.82	d	*D*	**D**	d	*D*	**D**	45.99	d	*D*	**D**	d	*D*	**D**	45.82	d	*D*	**D**	d	*D*	**D**
20	42.63	d	*D*	**D**	d	*D*	**D**	38.72	d	*D*	**D**	d	*D*	**D**	38.59	d	*D*	**D**	d	*D*	**D**
25	50.73	d	*C*	**C**	d	*D*	**D**	36.93	d	*D*	**D**	d	*D*	**D**	39.02	d	*D*	**D**	d	*D*	**D**
30	36.83	d	*C*	**C**	c	*C*	**C**	52.16	d	*D*	**D**	d	*D*	**C**	42.38	d	*D*	**C**	d	*D*	**D**
35	35.76	c	*C*	**C**	c	*C*	**C**	45.93	d	*D*	**C**	d	*D*	**C**	42.57	d	*D*	**C**	d	*D*	**D**
40	35.89	c	*C*	**C**	d	*C*	**C**	36.98	c	*C*	**C**	d	*D*	**C**	37.10	c	*C*	**C**	d	*D*	**D**
45	50.25	c	*C*	**C**	c	*C*	**C**	52.18	c	*C*	**C**	d	*D*	**C**	40.10	c	*C*	**C**	c	*D*	**C**
50	37.75	c	*C*	**C**	d	*C*	**C**	48.37	c	*C*	**C**	d	*D*	**C**	52.52	c	*C*	**C**	c	*C*	**C**
55	41.52	c	*C*	**C**	c	*C*	**C**	49.65	c	*C*	**C**	d	*D*	**C**	46.96	c	*C*	**C**	c	*C*	**C**
60	52.55	c	*C*	**C**	d	*C*	**C**	48.65	c	*C*	**C**	d	*D*	**C**	53.55	c	*C*	**C**	c	*C*	**C**
65	37.54	c	*C*	**C**	c	*C*	**C**	47.33	c	*C*	**C**	d	*D*	**C**	35.61	c	*C*	**C**	c	*C*	**C**
70	50.23	c	*C*	**C**	d	*C*	**C**	53.31	c	*C*	**C**	d	*D*	**C**	36.52	c	*C*	**C**	c	*C*	**C**
75	36.89	c	*C*	**C**	c	*C*	**C**	42.92	c	*C*	**C**	d	*D*	**C**	54.91	c	*C*	**C**	c	*C*	**C**
80	42.20	c	*C*	**C**	d	*C*	**C**	41.64	c	*C*	**C**	d	*D*	**C**	43.67	c	*C*	**C**	c	*C*	**C**
85	35.29	c	*C*	**C**	c	*C*	**C**	45.22	c	*C*	**C**	d	*D*	**C**	40.82	c	*C*	**C**	c	*C*	**C**
90	45.15	c	*C*	**C**	d	*C*	**C**	40.95	c	*C*	**C**	d	*D*	**C**	38.01	c	*C*	**C**	c	*C*	**C**
95	51.91	c	*C*	**C**	d	*C*	**C**	41.28	c	*C*	**C**	d	*D*	**C**	39.20	c	*C*	**C**	c	*C*	**C**
100	50.03	c	*C*	**C**	c	*C*	**C**	42.99	c	*C*	**C**	d	*D*	**C**	47.10	c	*C*	**C**	c	*C*	**C**

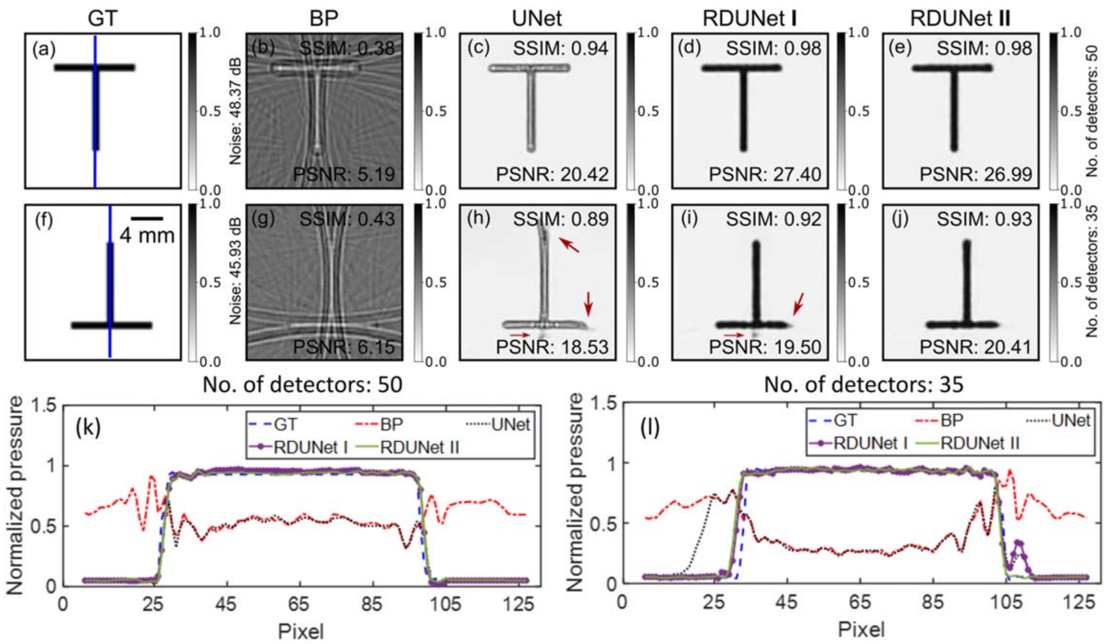

Figure 9. Same as figure 8 but for the T-shape phantom. Undesirable spots and/or regions with incomplete reconstructions are marked by the red arrows.

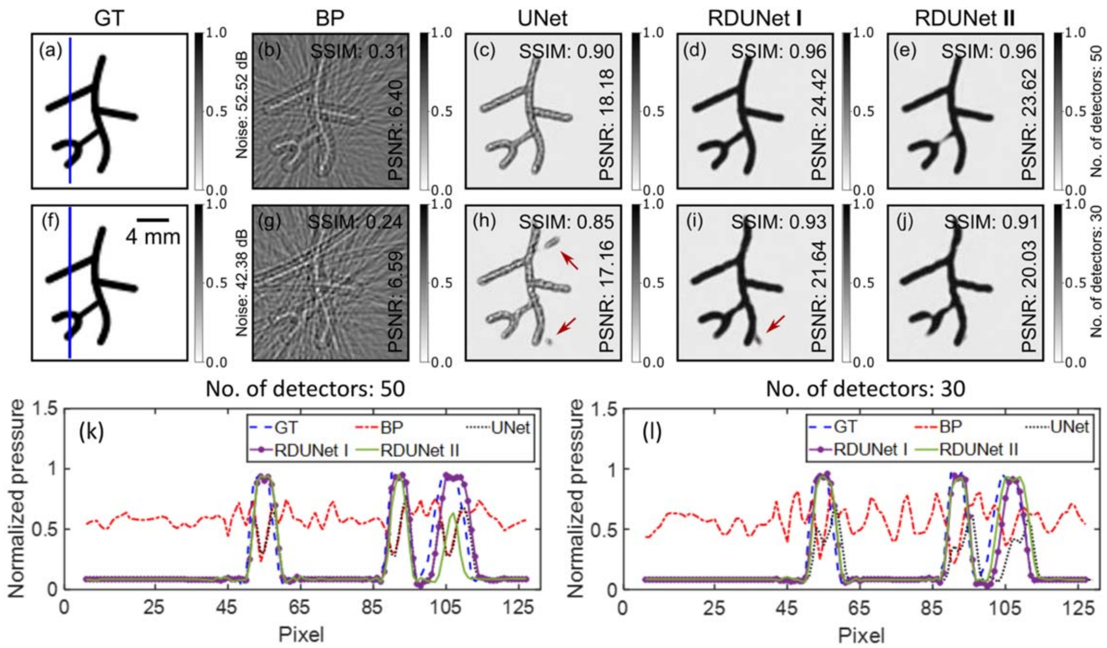

Figure 10. Same as figure 8 but for the vasculature phantom. The red arrows point irrelevant features.

reconstructed image prompted us to extend our investigation on experimental PA images. Figure 11 shows the photographs of the normalized reconstruction results using various algorithms at 50 and 30 detectors. The photographs of the sample (gold standard) at various positions are pasted on figures 11(a) and (f) as a ready reference. The BP images (normalized by the respective maximum values) differ greatly from the respective GTs as illustrated in figures 11(b) and (g). The shapes are also inaccurately reproduced by the UNet model (third column). Conversely, RDUNet I and II produce a faithful reconstruction of the corresponding GT. Figures 11(k) and (l) exhibits image profiles for the above two rows, respectively. Performance comparison of the reconstruction protocols can be made from these figures; table 2 offers a quantitative assessment [see table S1 (rows 5 to 23, columns 10 to 17) for details], while detailed qualitative results of these CNN frameworks are presented in table 3. Notably, RDUNet II consistently demonstrates excellent results in both cases.

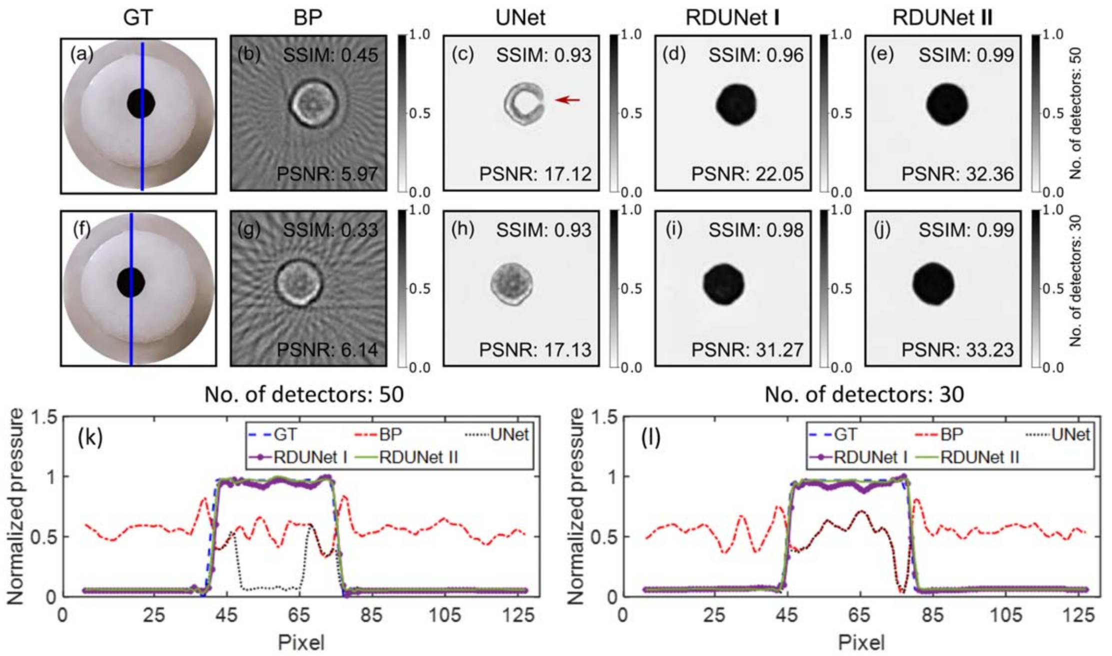

Figure 11. Image reconstruction by the BP and DL algorithms exploring the experimental dataset for the single-disk phantom. (a) and (f) Present GT images. (b)-(e) Outputs of different algorithms when 50 sensors are used to acquire the PA signals. (g)-(j) The PA signals are recorded by 30 detectors. The defective region is marked by the red arrow. (k) and (l) display image profiles for the top and middle rows, respectively. Line plots are generated along the straight lines demonstrated in (a) and (f). respectively.

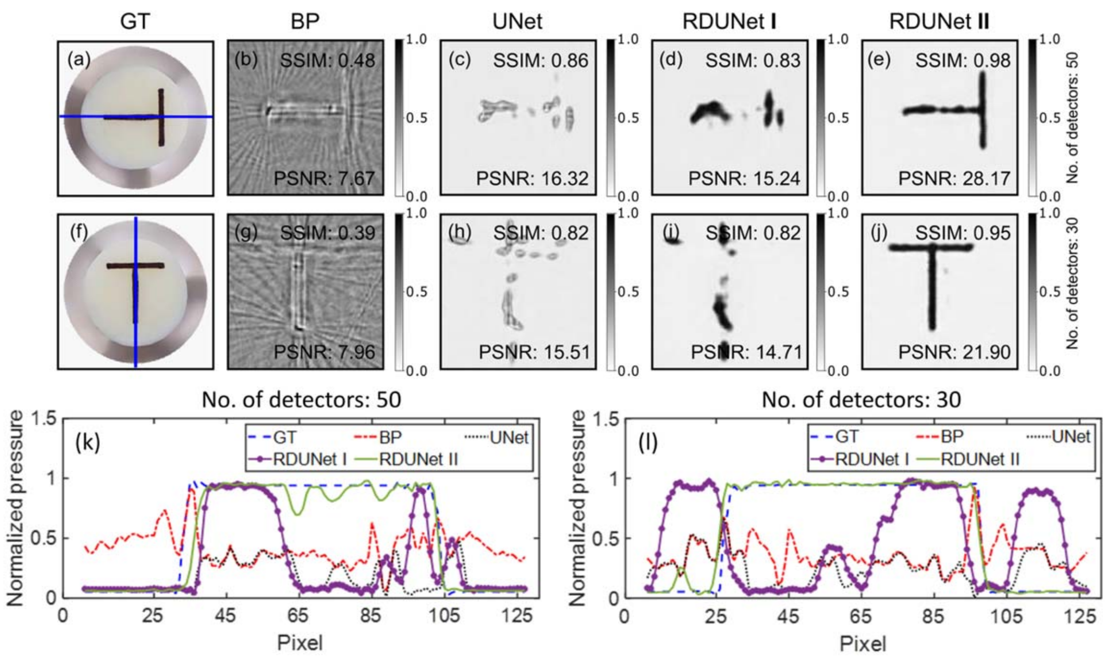

Figure 12. Same as figure 11 but for the T-shape phantom.

Moving forward, the findings for the T-shape experimental phantom are visualized in figure 12. It is observed that the outcomes for the BP algorithm are not convincing with less number of detectors. Moreover, both UNet and RDUNet I models struggle to produce faithful reconstructions. However, RDUNet II succeeds in forming the light-absorbing structures. This can be observed in figures 12(k) and (l) as well. Both qualitative and quantitative assessments further confirm the superior performance of the RDUNet II, highlighting the advantages of training it with a combination of simulated and experimental images [see table 2, table S2 (rows 5 to 23, columns 10 to 17), table 3].

Finally, figure 13 demonstrates the ability of faithful reconstruction of a vasculature phantom using

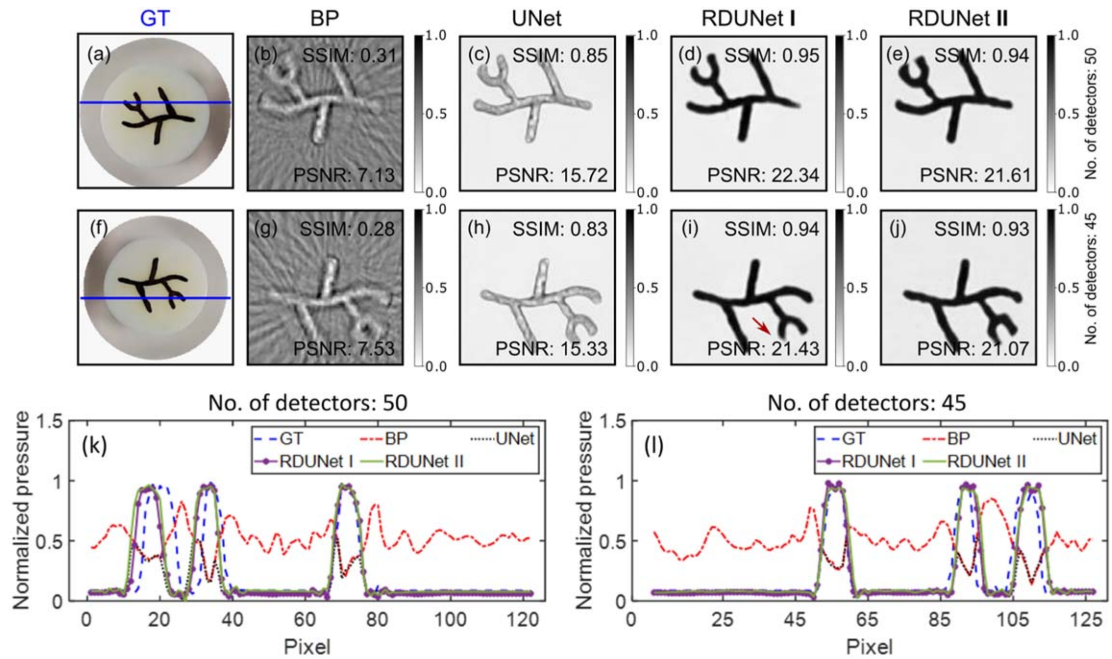

Figure 13. Same as figure 11 but for the vasculature phantom. The red arrow points either defective or incomplete reconstruction.

various reconstruction algorithms. As expected, the BP algorithm is unable to perform good results with a lesser number of USDs. It can be noticed from figure 13(c) and (h) that UNet can produce convincing results for 50 and 45 sensors, however, the branches of the vasculature phantom appear greyish and imperfectly visible. Additionally, RDUNet I demonstrates faithful performance for 50 sensors, with a slight incomplete reconstruction in the Y-shaped bottom branch for 45 sensors. The RDUNet II provides a satisfactory performance [see figure 13(e) and (j)]. As metnonted above, the corresponding image profiles are plotted in figures 13(k) and (l). The outcomes of quantitative and qualitative evaluation reveal that RDUNet I and II outperform BP and UNet in terms of performance [refer to table 2, table S3 (rows 5 to 23, columns 10 to 17), table 3].

4. Discussion and conclusions

Medical imaging has been widely used in clinics to help doctors get a better diagnosis and deliver optimal medical care. As a result, producing high-quality medical images is crucial for successful medical interventions and the image reconstruction algorithm plays a pivotal role in achieving this objective. As mentioned earlier, there are primarily two techniques that exist for image reconstruction: analytical methods and model-based approaches. Each one presents its advantages and limitations. The first method represents the more straightforward theoretical approach among reconstruction methods. Although it offers computational efficiency and speed but may falter in situations where complete data acquisition is not feasible. Full-view datasets can pose challenges in real-time imaging scenarios, where incomplete or noisy data may be recorded. Conversely, the iterative approaches perform well with limited view data and provides accurate reconstruction. For example, the compressed sensing technique implementing l_1 regularization method can efficiently recover the ground truth utilizing sparsely sampled dataset [18, 20]. Note that solving the l_1 regularization problem does not impose significant computational burden but building a system matrix is a time consuming task and a high performance computer is required to handle such a large matrix [13, 14]. Therefore, compressed sensing procedure may be suitable for rapid imaging applications if the system matrix is known a-priori. For the iterative approaches, various imaging degrading factors (originating from instrument and medium dependent properties), if incorporated within the system matrix, get canceled out while calculating the solution and thus facilitates faithful image reconstruction.

In PAT, a single-element detector circularly rotates around the sample to capture the PA signals for in-vivo image formation using various reconstruction algorithms. In this context, the BP algorithm has been employed to reconstruct these images. However, due to limitations in the storage capacity of the DAQ card and the need for rapid image formation, PA signals are often captured at limited detector locations. As a result, the BP algorithm produces artifacts in the reconstructed images, making them less clear and reducing their quality. Here we have applied a DL-based CNN algorithm to mitigate this problem. It may be pointed out that a DL network is trained using a dataset and is not programmed. It learns salient features from the dataset and therefore does not need expert analysis in order to achieve improved performance. Moreover, the same network can be retrained with a different dataset to attain different goals. On ther hand, conventional image processing algorithms involve a few lines of codes and may become effective in many situations [54]. However, manual intervention is required to tweak parameters of a conventional algorithm to achieve better performance.

DL has emerged as the state-of-the-art ML method for various applications of PA imaging. A DL-based CNN framework incorporates both pre-processing and post-processing approaches to strengthen image quality and remove artifacts. In pre-processing, PA signals at various locations can be predicted; however, subsequent reconstruction algorithms may reintroduce errors. In this paper, we have proposed a post-processing-based RDUNet model to alleviate the image degradation arising from a restricted number of USDs. The RDUNet framework builds on two following methodologies:

- Replacing standard convolution, max-pooling operations with densely connected convolutional layers to reuse the feature maps within the encoder and decoder sections.
- A local residual learning is added by shortcut connections to each dense block to avert the gradient disappearance in the overall neural network and accelerate the learning process.

The performance (qualitatively and quantitatively) of the RDUNet and comparison with other reconstruction algorithms are illustrated in table 2 and 3. It is important to note that the a DL network takes lesser time (testing time) to generate an image than that of the BP algorithm (analytical algorithm); see the last column of table 2.

In this work, the RDUNet was trained with two types of datasets. In the first protocol, the RDUNet was trained with 1665 heterogeneous simulated images of three different kinds of phantoms, and the trained network was termed as the RDUNet I. Subsequently, its efficacy was scrutinized with both simulated and experimental images. The model showcases accurate reconstruction of simulated images with up to 25 detectors for single-disk images and 40 detectors for T-shape and vasculature images. Additionally, the network exhibits successful reconstruction of experimental single-disk and vasculature phantoms for up to 30 and 50 detectors, respectively. However, the network struggles to produce the reconstructed T-shape images regardless of the number of detectors employed. In general, the number of optimal detectors for precise image reconstruction increases by approximately 60% as we move from simple to complex targets (single-disk to vasculature).

Consequently, we trained our RDUNet with the second type of dataset. The second dataset was a mixture of simulated (1665) and experimental (405) images of three analogous phantoms, which served as an opportunity for the RDUNet II model to enhance adaptability to a wider range of experimental challenges. The results show that RDUNet II can produce accurate reconstructed images up to 25, 35, and 30 detectors for simulated single-disk, T-shape, and vasculature images, respectively. For experimental single-disk and vasculature phantoms, the model accomplishes acceptable reconstruction with 30 and 45 detectors, respectively. Notably, RDUNet II outperformed RDUNet I in T-shape phantom reconstruction, achieving accurate reconstruction with up to 30 detectors. It may be noted that 20 and 50% more detectors are needed for concise reconstruction of vasculature target compared to the single-disk structure for simulated and experimental cases, respectively).

In this work, the DL methods have been deployed for image reconstruction at various noise levels. Electronic and thermal noise dominate in PAT imaging. The noise level associated with a PA signal would depend on the choice of the bandwidth and bandwidth distribution of the sensor. Due to the presence of noise, it may become difficult to accurately retrieve the quantitative information of the physical parameter. It is known from the literature that a BP image retains low frequency components of the GT and thus exhibits image blurring [55]. However, our results show that the RDUNet II algorithm can reproduce the GTs very well. Particularly, edges are clearly replicated. Therefore, it can be speculated that the RDUNet II algorithm faithfully preserves the frequency components present in the GT images.

Even with many promising outcomes from the study, there are several issues yet to be resolved. First, efforts may be directed to further reduce the number of detectors without compromising the accuracy of the RDUNet on the heterogeneous dataset by incorporating a DL-based classification model as discussed in [31]. Secondly, we need to investigate the adaptability of the RDUNet when a full-view PA dataset is unavailable. The performance of these DL based methods, applied herein, needs to be compared with the recently developed approaches. Lastly, the performance of this network may be assessed with in-vivo datasets.

In conclusion, this work indicates the potential of integrating DL for enhancing PAT reconstruction quality with minimal number of USDs. This is not only lowers the hardware complexity and cost but also suppresses the time required for data acquisition, which is crucial for real-time imaging, associated with drug delivery monitoring, tumor diagnosis, and other biomedical imaging tasks. We compare RDUNet I and II with the traditional BP algorithm and a predesigned UNet using simulated and experimental phantoms (single-disk, T-shape, vasculature). It is found that for simple shapes acceptable images can be formed using sensor data acquired by approximately 30 USDs with the help of RDUNet II for full view 2D PAT; whereas PA signals captured at $\approx$40 angular locations are required for faithful reconstruction of the vasculature phantom by the same algorithm. Moreover, the requirement of the optimal number of detectors for RDUNet II is 10% less compared to that of RDUNet I for accurately reconstructing the vasculature phantom. The result can prove the effectiveness of the proposed framework and shows that the RDU-Net II (i.e., the DL network trained with a combination of the simulated and experimental datasets) can help to reproduce a faithful PAT image with a significantly lesser number of USDs.

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Author contributions

SM, PW, ZK, and RKS performed the numerical simulations. SM and NS designed the deep learning model. SM and SP performed phantom experiments and analyzed the data. SM prepared the figures. SM and SP drafted the manuscript. RKS and NS reviewed and edited the manuscript. All authors read and approved the submitted manuscript.

Conflicts of interest

The authors declare no conflicts of interest.

Data availability statement

The data cannot be made publicly available upon publication because they are not available in a format that is sufficiently accessible or reusable by other researchers. The data that support the findings of this study are available upon reasonable request from the authors.

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